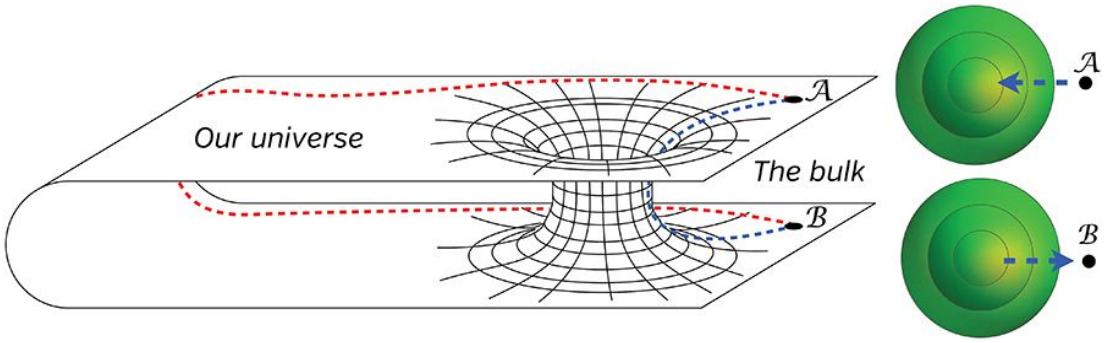


Fig. 14.2. Flamm's wormhole.

Wormhole Collapse

It is often difficult to extract, from the mathematics of Einstein's equations, a full understanding of their predictions. Flamm's wormhole is a remarkable example. From 1916 until 1962, nearly a half century, physicists thought that the wormhole is static, forever unchanging. Then John Wheeler and his student Robert Fuller discovered otherwise. Looking much more closely at the mathematics, they discovered that the wormhole is born, expands, contracts, and dies, as shown in Figure 14.3.

Initially, in picture (a), our universe has two singularities. As time passes, the singularities reach out to each other through the bulk and meet to create the wormhole (b). The wormhole expands in circumference, (c) and (d), then shrinks and pinches off (e), leaving behind the two singularities (f). The birth, expansion, shrinkage, and pinch-off happen so quickly that nothing, not even light, has time to travel through the wormhole from one side to the other. Anything or anyone that attempts the trip will get destroyed in the pinch-off!